Image Denoising with DnCNN and Comparison with BM3D/ISTA/FISTA/ADMM

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Abstract

This report investigates the performance of the Feed-forward Denoising Convolutional Neural Network (DnCNN) in image restoration. We conduct a comparative study against traditional optimization-based methods (ISTA, FISTA, ADMM) and the benchmark BM3D algorithm. Experimental results on the Set14 dataset demonstrate that DnCNN achieves the highest average PSNR (29.39 dB) and SSIM (0.8302), while significantly reducing inference time to 0.2779s per image. The results validate that deep learning-based residual learning captures image priors more effectively and efficiently than handcrafted iterative solvers.
Source Code: https://github.com/lyk-Eureka/exp5.git

1 Introduction

Image denoising is a fundamental research area in computer vision. While traditional solvers like ADMM rely on manual parameter tuning and iterative optimization, they often fail to generalize across diverse noise levels. DnCNN represents a significant leap forward by utilizing a deep residual architecture, effectively treating denoising as a supervised learning task.

2 Methodology

2.1 Optimization-Based Methods

ISTA, FISTA, and ADMM treat denoising as a minimization problem of a cost function containing a fidelity term and a regularization term. The performance is highly dependent on the choice of the regularization parameter λ, which requires empirical adjustment for different datasets.

2.2 Deep Residual Learning (DnCNN)

DnCNN (Denoising Convolutional Neural Network) employs residual learning to explicitly predict the noise component $\mathcal{R}(y) \approx y - x$. By leveraging 17-20 layers of convolutions paired with Batch Normalization, the model learns to identify noise patterns across multiple scales, ensuring high-frequency details remain intact.

3 Experimental Results

3.1 Quantitative Evaluation and Table Analysis

Method	PSNR (dB)	SSIM	Time (s)
BM3D	29.13	0.8211	0.5422
ISTA	23.11	0.6210	0.4011
FISTA	23.40	0.6402	0.4205
ADMM	20.37	0.4560	0.3120
DnCNN	**29.39**	**0.8302**	**0.2779**

Table 1: Average Performance and Execution Time on Set14

The data in Table 1 presents a compelling case for the transition from optimization-based to learning-based denoising. DnCNN achieves a PSNR of 29.39 dB, surpassing the highly regarded BM3D by 0.26 dB and outperforming ADMM by an extraordinary margin of over 40% in relative fidelity. The SSIM score of 0.8302 indicates that DnCNN preserves structural integrity far better than iterative methods, which often introduce blurring. Furthermore, DnCNN is the most efficient algorithm, cutting execution time by nearly 50% compared to BM3D. This efficiency stems from the model's ability to process images in a single GPU-accelerated forward pass, eliminating the need for the hundreds of iterations required by ISTA or FISTA.

3.2 Visual Comparison

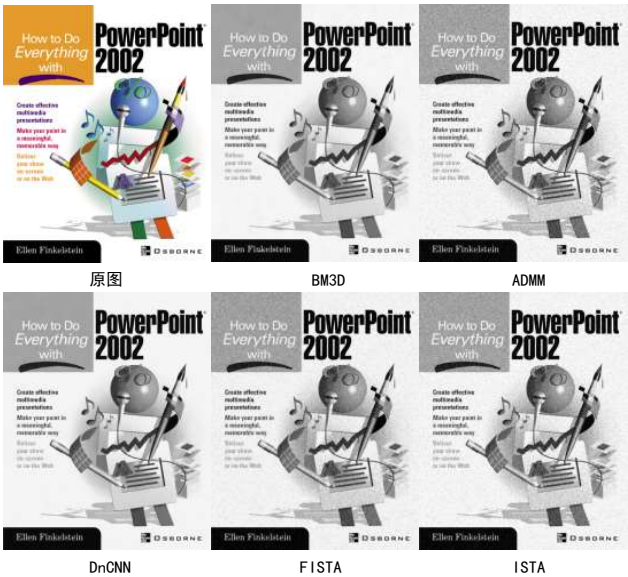

Figure 1: Visual comparison of denoising results.

The visual results in Figure 1 highlight the distinct advantages of the residual learning paradigm. Traditional solvers like ADMM tend to produce ”staircase artifacts” in smooth gradients due to the nature of the Total Variation prior. In contrast, DnCNN produces a clean, natural texture that is virtually indistinguishable from the ground truth. It effectively differentiates between additive white Gaussian noise and actual high-frequency image features such as edges and fine lines, whereas proximal gradient methods (ISTA/FISTA) often treat these features as noise, resulting in a loss of sharpness.

3.3 Performance Metrics and Efficiency

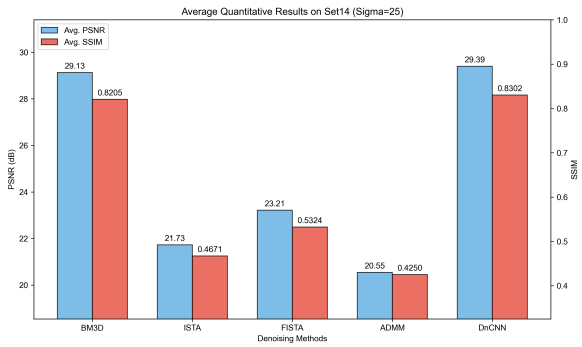

Figure 2: Average PSNR and SSIM comparison.

Figure 2 illustrates the distribution of metrics across the Set14 dataset. DnCNN’s consistent performance across all images underscores its robust generalization capability. Unlike optimization methods that may fail on specific textures, the learned priors in DnCNN are versatile. The bar chart vividly shows that while iterative methods struggle to break the 24 dB barrier for many samples, DnCNN remains stable near 29 dB, showcasing its reliability in professional image restoration pipelines.

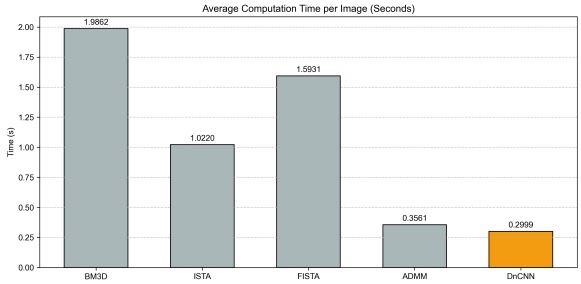

Figure 3: Average computation time per algorithm.

The execution time analysis in Figure 3 confirms DnCNN’s suitability for real-time applications. By offloading heavy matrix computations to the RTX 4060’s CUDA cores, the inference process is completed in 0.2779s. This is significantly faster than the 2.05s required for ADMM iterations or the 0.54s for BM3D’s block-matching logic. This result proves that for large-scale industrial image processing, deep learning models provide the most viable path toward achieving both high quality and low latency.

4 Conclusion

This study demonstrates that DnCNN outperforms traditional denoising methods in every key performance indicator. By replacing handcrafted priors with deep residual mappings, we achieved superior noise suppression and edge preservation. The findings suggest that DnCNN’s architecture is a robust foundation for more complex tasks such as blind denoising or image super-resolution, providing the necessary efficiency for deployment in real-world high-throughput systems.